

The Xeon vs the Itanium		
	Xeon	Itanium
Available Speeds	1.40 GHz, 1.50 GHz 1.70 GHz	733 MHz and 800 MHz
Features	Intel® Net Burst™ Micro architecture 400 MHz Front Side Bus Hyper-Pipelined Technology Execution Trace Cache Rapid Execution Engine Streaming SIMD Extensions 2 (SSE2) Advanced Dynamic Execution Double Precision Floating Point RISC based processor	IA-64 400 MHz Front Side Bus 64-bit addressing and high-memory band- width for superior memory scaling Transaction processing and large memory for enterprise solutions Data integrity via machine check architec- ture, ECC and parity RISC based processor with EPIC functionality.
Chipset	Intel 860 Chipset	Intel 460GX Chipset
Cache	L1 Execution Trace Cache (8 Kb + 12 KB) L2 Advanced Transfer Cache - 256 KB	L1 (128 KB) L2 (256 to 1 MB) L3 (2 MB to 4 MB)
RAM	Dual Channel RDRAM	SDRAM and DDR SDRAM
Fabrication Process	0.18 micron process	0.18 micron process
Transistor Count	42 million	25.4 million

information and can also process data, 64 bits at a time. This is precisely where the Itanium scores over other processors that are built around a 32-bit architecture. A 32-bit processor can theoretically access approximately 4.3 billion bits of data per second, whereas the Itanium can access around 2.1 billion GB (18.4 quintillion bits), that is, effectively 4 billion times more than the current 32-bit processors!

Enhanced floating point: Another advantage with this processor is that it has a huge number of registers. Registers are the areas within the processor where instructions act on data. The Itanium has 128 82-bit floating-point registers, 64 1-bit and 64 8-bit Predicate and Branch registers. Due to the 128 registers of the Floating Point Unit, it is ideally suited for executing complex tasks and computations that specifically make use of this unit, such as weather predictions, industrial level 3D modelling, etc.

Multimedia architecture: Multimedia applications require more CPU power and RAM. The Itanium is ideal for running multimedia applications due to its capability of executing arithmetic and data arrangement functions in parallel.

A new chipset: As is the case with all new processors, even this processor will have a new motherboard with a Slot M or C4 (flip chip assembly) based on the all-new AL 460 GX chipset which can support from one to four Itanium processors and up to 64 GB of RAM. Another chipset, BS 460 GX, will support one or two Itanium processors. This chipset supports PC100 SDRAM (Synchronous Dynamic RAM) and PC1600 DDR SDRAM only. The reason for using SDRAM is that it is extremely cost-effective and since high-end servers and workstations use large amounts of memory, RDRAM

(Rambus Dynamic RAM) would greatly increase the overall system cost.

Compatibility: According to Intel, Itanium provides full compatibility with 32-bit applications, implying that all current 32-bit based applications will run on it without any modifications. However, 32-bit applications will not run faster on an Itanium. In fact, there are chances that there would be no noticeable speed difference between, say, a Coppermine 800 MHz and an Itanium 800 MHz. While a Coppermine is tuned to run 32-bit applications, the Itanium is built to run 64-bit applications and hence would not be able to run 32-bit applications as fast as a 32-bit processor. An ideal condition would be an Itanium processor running on a 64-bit operating system with a target application that is also 64-bit based.

Soon, but not yet

Time and experience have shown that whenever a new processor is launched, its true power can only be experienced a year or two after its launch, when applications that actually tap its potential are made available. Hence, you may not find a distinct difference while using newer processors with older applications. The same would be the case with the Itanium—the power of this processor will come to the fore only when 64-bit operating systems and applications are made available.

The Itanium would be suitable for large corporates and enterprises that require a high level of computing power and need to ‘future-proof’ their investments in new servers. As it stands, 64-bit is the way processors are headed and for an enterprise that uses high performance applications, the Itanium would prove to be a good investment. 📌

Competition for Itanium

Sledgehammer is AMD’s most ambitious project to date. The chip will run software designed for 32-bit processors, such as Intel’s Pentium III and AMD’s Athlon, as well as software for 64-bit chips, a class of processors that includes Sun Microsystems’ Ultra Sparc, Compaq Computer’s Alpha and Intel’s upcoming Itanium.

Sledgehammer will use the same basic instruction set as the Athlon but will contain features that allow the chip to run 64-bit programs. As a result, software makers only need to revamp their programs, and not write completely new ones as in the case of Itanium.

The x86-64 platform will offer two distinct modes of operation: long and legacy modes. Long mode provides support for all 64-bit operations and allows support for existing 16/32-bit application execution through a long mode compatibility option. To operate in either form of long mode, the system must be utilising a 64-bit compliant operating system. By offering two levels of long mode support for all 16-, 32-, and 64-bit x86 applications, AMD is trying to position itself as the leader in performance, support, and compatibility.

The Sledgehammer will be running at 2 GHz with 0.13 micron and with a die size of 105 mm² with 266 FSB.